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What is the Ethereum Ice Age?

11 comments | Antonio Madeira | 28 Jul 2016 | 51.85 k

Ethereum is currently a Proof of Work cryptocurrency, meaning that computational power is needed, not only to produce new coins but to process transactions and to keep the entire ecosystem moving. In order to ensure the system is scalable and decentralized, Ethereum plans to move to a Proof of Stake protocol where a lot less computational power is required and miners can earn rewards according to their balance, a bit like gambling but without the risk, so if that's what you're looking for, we`ve created the Ethereum casino sites in 2020. The Ethereum Foundation is still working on the Proof of Stake protocol, Casper.

Since Casper is not finished and it's not contained in the release version of Ethereum a hard-fork might be required to implement this change. An hard-fork creates an incompatibility between the previous version and the latest, there is always the possibility to create a "split" that result in two blockchains, like Ethereum Classic, did on the 1920000th block, after the hard-fork to refund The DAO token holders took place.

In order to ensure such event doesn't take place (although it did already) and to give themselves a time-frame to finish Casper (making the community aware of the introduction of a hard fork within that time-frame), a Difficulty Time Bomb, is also known as Ice Age was implemented in Ethereum.

Part 1 - What is the Ethereum Ice Age

The Ethereum Ice Age is a difficulty adjustment scheme that was put in place to ensure that everyone has an incentive to move to the new blockchain once the hard-fork is implemented. It was introduced on the 7th of September (2015-09-07), about 11 months ago and it's programmed to raise difficulty exponentially.

Click here to see a difficulty simulation of the increase in difficulty.

It's impossible for miners to keep up with the increase of difficulty which would raise block time and it would make the blockchain freeze, hence the name Ice Age.

Part 2 - Resetting difficulty

The key part is the calculation for the increase in difficulty is Math.pow(2,Math.floor(block.number / 100000) - 2). The current hash rate can deal with it as the block number is 2 million so the calculation works out at 262,144 and the current difficulty is around $5*10^{13}$

By block 4.8 million we should see bigger numbers than the current difficulty - Math.pow(2,Math.floor(4800000 / 100000) - 2) = $7*10^{13}$

As it's taken around 1 year for 2 million blocks, we should have another 1 year and 5 months until we feel the effect of the 'Time Bomb';

This means that in 16-17 months, if the hashing power stays the same, we should start feeling the effect of this time bomb. Ethereum will be forced to either introduce Proof of Stake, to remove this feature or to reset "the clock" on the difficulty bomb.

Hopefully Casper, the Proof of Stake Protocol, will be ready otherwise a hard fork will be need in order to either remove the Ice Age feature. This would give developers unlimited time to work on Casper, or to delay the Difficulty Bomb. But we have a lot of time to do so.

A better explanation can be found here, thanks benjaminion for your help.

It's unclear which one of these options, or even if a third one, will be chosen by the developers. CryptoCompare will update this article as soon as more information is available.

Comments

What is your opinion on What is the Ethereum Ice Age? (You will be banned if you post affiliate / referral links. Please keep the conversation civilized and remember, we're all in this together!)

Post

Coreol
6 years ago

What I still don't understand, is how can the system still "work" when there's no more proof of work? Transactions will still take place and smart contracts will exist - which will need "power" to run. Who's going to provide the computational power if PoS will make "mining" as it is done today obsolete? What am I missing here?

View Edits

4 replies | Agree

mike_blue7
6 years ago

great info. good to know. thanks Vlad.

reply | Agree 1

OldBird
6 years ago

For me, i will get out of any classic PoS situation regardless. I wont have enough of a stake to provide anything besides a very minuscule amount of reward and that investment would be better put in future coins

2 replies | Agree 1

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Dev team is solely dedicated to ETH I'm failing to see how ETC can escape the brought in to remove the timer.

I course that's a few minutes of thought and the topic requires more research on my part. Just getting a little annoyed with Ethereum when XEM, NXT, FCT, etc have projects I'm also trying to keep up to date on.

4 replies | Agree 1

rushdmowjee
6 years ago

How come we weren't told about this earlier. As a miner(and I'm sure there are lots of other miners out there), who kept trying to get information on when the difficulty would scale up but there was no sure shot answer and now we get info on the Ice age?! Ether needs to communicate better, they barely even up to date the blog.

2 replies | Agree 2

ZeroCool86
6 years ago

Any up to date info on this is appreciated. This is everything we could find so far.

reply | Agree

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